COMPUTER NETWORKS (CSA0711)
ASSIGNMENT 2 — CO2

CO2 Coverage: Configure IP addressing schemes, subnetting and basic routing techniques using a simulation tool, for efficient network design and topology – packet switching, IPv4 addressing, subnetting, distance-vector and link-state routing, multicast, ICMP, IPv6, software-defined networking, network virtualisation and OpenFlow.

Nature of Assignment: Individual
Duration: 1 Week
Total Marks: 50
Simulation Tool: Cisco Packet Tracer (or GNS3)
Submission Date: 12/11/2026
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Course Code: CSA0711
Course Name: Computer Networks
Bloom's Levels Assessed: K5 – Evaluate (every task asks you to calculate or analyse first, and then judge, decide or justify)

Mandatory Execution and GitHub Submission

Every student must individually execute every task in this assignment, including all "Complete the Code" programs.

Create a GitHub repository named `CSA07-<RegisterNumber>-Assignment2` and share it with the course faculty (public, or private with the faculty added as a collaborator).

Keep one folder for each task (for example: `Task01` , `Task02` … or `Q1` , `Q2` …), containing the Python programs, Packet Tracer (`.pkt`) files, device configurations, Wireshark captures, command outputs and screenshots.

Include a `README.md` with your name, register number, the operating system, Python version and Packet Tracer (or NS-3) version used, the steps to run each program or open each simulation, and a short note of the result of each task.

Commit your work progressively – at least one commit for each task. The commit history will be checked; a single upload of all files on the last day will not be accepted.

Submit the GitHub link below and on the cover page of your report:

How to Answer the Additional Questions

Complete the Code: Copy the given program or device configuration into your repository, replace every `TODO` with working code or commands without changing the rest, run it (Python 3 or Cisco Packet Tracer) and compare your output with the expected output or result.

Alternative Concept Analysis: Describe how the solution would change if the alternative concept were used, and evaluate both solutions using network measures such as throughput, delay, jitter, packet loss, overhead, reliability, security and cost, supported by your own calculations, simulations or measurements wherever possible.

Q1 — Spot the Invalid Addresses (6 Marks • Bloom's Level: K5 – Evaluate)

Problem Context – Address Clashes in a Hospital Group: A hospital group's IT team has collected the addresses configured on some devices. Several devices cannot communicate, and one clinic says it has "run out of addresses".

Admission PC 172.16.45.200 /20 gateway 172.16.32.1
Lab printer 10.0.15.255 /21 gateway 10.0.8.1
X-ray server 192.168.4.0 /23 gateway 192.168.4.1
Clinic router 203.0.113.10 /29 (public, from the internet provider)

Your Challenge: For each device, find the network address, broadcast address and usable host range, and decide whether the address is a valid host address. Two of them contain a trap. Inquiry: The clinic's 18 devices share the one public /29 address block through NAT. Show how the NAT table lets two PCs both use source port 50000 at the same time. Then explain why packets of the same file can arrive in a different order (datagram switching), and what would be different with virtual circuits.

Alternative Concept Analysis: Suppose the clinic were given a public /27 instead of using NAT behind a /29. Count the public addresses needed for 18 devices, state the cost and exposure to the internet, and identify which application problems NAT would no longer cause. Evidence Expected: Correct network, broadcast and host ranges, the two traps identified with reasons, a NAT table example and a reasoned comparison of datagram and virtual-circuit switching.

Q2 — Subnet a Campus That Keeps Growing (8 Marks • Bloom's Level: K5 – Evaluate)

Problem Context – Engineering College Campus: A college has `192.168.10.0/24` for laboratories (100 hosts), staff (50 hosts), the library (25 hosts) and a point-to-point WAN link (2 hosts).

Your Challenge: Design the subnets twice – with equal-size subnets and with VLSM – giving network, prefix, mask, first and last host and broadcast for each. Then answer the trap: six months later the college adds a hostel network of 60 hosts. Can it be fitted into the remaining space of your VLSM plan without renumbering? If not, what is the smallest change that makes it fit? Route Summarisation: The city campus uses `172.16.8.0/24` , `172.16.9.0/24` , `172.16.10.0/24` and `172.16.11.0/24` . Find the single summary route, and show what goes wrong if `172.16.12.0/24` is added later and someone summarises it as `172.16.8.0/21` .

Complete the Code: Complete `TODO 1` below, and run the program with Python 3. Check every subnet with Python's `ipaddress` module, then add the 60-host hostel network and see what happens.

```
"""Variable-length subnetting (VLSM) of 192.168.10.0/24 for a college campus.""" import math
def to_int(ip):
    a, b, c, d = map(int, ip.split("."))
    return (a << 24) | (b << 16) | (c << 8) | d
def to_ip(n):
    return ".".join(str((n >> s) & 255) for s in (24, 16, 8, 0))
def vlsm(base, needs):
    addr = to_int(base)
    for name, hosts in sorted(needs, key=lambda x: -x[1]):
        # TODO 1: Find the smallest block that holds hosts + 2 addresses (a power of # two), its prefix length 32 - log2(block), and print the network, prefix, # first and last usable host and broadcast address. Then move addr to the next # block
        pass # replace this line with your code
    return to_ip(addr)
```

```
needs = [("Labs", 100), ("Staff", 50), ("Library", 25), ("WAN link", 2)] print("next free address:",
vlsm("192.168.10.0", needs))
```

Alternative Concept Analysis: Redo the college plan starting from the smallest subnet instead of the largest. Show where the allocation goes wrong (overlapping or misaligned blocks) and explain why VLSM must assign the largest subnets first.

Evidence Expected: Complete subnet tables for both methods, a correct answer to the hostel trap, a correct summary route and an explained summarisation error, verified in Packet Tracer.

Q3 — Which Way Will the Packet Go? (6 Marks • Bloom's Level: K5 – Evaluate)

Problem Context – Two Campuses and a Backup Link: The main-campus router has this routing table:

```
S 10.0.0.0/8 via 192.0.2.1
S 10.1.0.0/16 via 192.0.2.5
S 10.1.2.0/24 via 192.0.2.9
S 10.1.2.128/25 via 192.0.2.13
S* 0.0.0.0/0 via 198.51.100.1
```

Your Challenge: Determine the next hop for packets to `10.1.2.200` , `10.1.2.100` , `10.1.9.9` , `10.7.7.7` and `8.8.8.8` , and state the rule you used. Then build the two-campus network in Packet Tracer with static routes, add a backup link with a floating static route (administrative distance 200), and show that traffic moves to the backup when the main link is shut down. Complete the Code: Complete `TODO 1` and `TODO 2` in the router configuration below, and enter it on R1 in Cisco Packet Tracer. Configure R2 in the same way with the reverse route. Remove the clock rate on the DCE end and record what happens to the serial link.

```
! Cisco Packet Tracer: static routing between the Main campus (R1) and City campus (R2). ! R1 LAN 192.168.10.0/24 on
G0/0, R2 LAN 192.168.20.0/24, serial link 10.0.0.0/30. enable
configure terminal
hostname R1
interface GigabitEthernet0/0
  ip address 192.168.10.1 255.255.255.0
  no shutdown
! TODO 1: Configure Serial0/0/0 with 10.0.0.1/30, set the clock rate to 64000 (R1 is the ! DCE end) and bring it up
exit
! TODO 2: Add a static route to the City campus LAN 192.168.20.0/24 through the next hop ! 10.0.0.2
end
write memory
show ip route
```

Alternative Concept Analysis: Replace the five static routes with OSPF on both campuses. Count the configuration lines needed for 2 campuses and for 5 campuses in each approach, and compare the time to recover when the main link fails.

Evidence Expected: Correct next hops with the longest-prefix rule, a working Packet Tracer network with a floating static route, and before-and-after routing tables.

Q4 — Watch Distance-Vector Routing Count to Infinity (6 Marks • Bloom's Level: K5 – Evaluate)

Problem Context – District Government Offices: Five routers connect government offices with the link costs A–B 2, A–C 7, B–C 3, B–D 8, C–D 1, D–E 4.

Your Challenge: Build every router's distance table round by round until nothing changes, and state how many rounds it took. Explain why A's best route to C is not the direct link. Now take three routers in a line X–Y–Z (cost 1 each). The link Y–Z fails. Show, round by round, how X and Y keep raising their distance to Z (count to infinity), and how split horizon with poison reverse stops it. Configure RIP in Packet Tracer and observe the convergence after a link failure. Complete the Code: Complete `TODO 1` below, and run the program with Python 3. After convergence, set the cost of link C–D to 99 (failed) and rerun; record the new tables and rounds.

```
"""Distance-vector routing (Bellman-Ford exchange) among five campus routers.""" INF = 99
names = "ABCDE"
cost = {("A", "B"): 2, ("A", "C"): 7, ("B", "C"): 3, ("B", "D"): 8,
        ("C", "D"): 1, ("D", "E"): 4}
link = {}
for (u, v), c in cost.items():
    link[u, v] = link[v, u] = c
table = {u: {v: (0 if u == v else link.get((u, v), INF)) for v in names} for u in names} rounds = 0
changed = True
while changed:
    changed = False
    rounds += 1
    old = {u: dict(table[u]) for u in names} # vectors received this round for u in names:
        for n in names:
            if (u, n) not in link:
                continue
            for dest in names:
                # TODO 1: Bellman-Ford update: if going through neighbour n (link cost + # n's old distance to dest)
                is cheaper than table[u][dest], update it # and set changed = True
                pass # replace this line with your code
print(f"converged after {rounds} rounds")
for u in names:
    print(u, " ".join(f"{table[u][v]:2d}" for v in names))
```

Alternative Concept Analysis: Run the same five-router network with link-state routing instead. After the C–D link fails, compare how many messages each method sends and how many rounds pass before every router has a correct route.

Evidence Expected: Correct round-by-round tables, a traced count-to-infinity example, the effect of poison reverse and RIP evidence from Packet Tracer.

Q5 — Why Does OSPF Treat 1 Gbps and 10 Gbps the Same? (6 Marks • Bloom's Level: K5 – Evaluate)

Problem Context – A State Internet Service Provider: An internet service provider connects cities with 10 Mbps, 100 Mbps, 1 Gbps and 10 Gbps links. After moving to OSPF, traffic still uses a 1 Gbps path instead of a parallel 10 Gbps path.

Your Challenge: Calculate the OSPF cost of each link speed with the default reference bandwidth of 100 Mbps, and explain the problem. Choose a new reference bandwidth, recalculate the costs, and run Dijkstra's algorithm by hand from one router on a 6-router network of your design. Configure OSPF in Packet Tracer (with `auto-cost reference-bandwidth`) and show that the 10 Gbps path is now chosen.

Decide whether the provider should use one OSPF area or several, and justify the decision.

Alternative Concept Analysis: The provider could use EIGRP (which uses bandwidth and delay in its metric) or IS-IS instead of OSPF. Compute the metric each would give the 1 Gbps and 10 Gbps paths, and decide which protocol avoids the problem without changing the reference bandwidth.

Evidence Expected: Correct cost calculations, a hand-worked shortest-path tree, Packet Tracer evidence and a justified area design.

Q6 — IPv6, ICMP, Multicast and SDN in a Smart City (5 Marks • Bloom's Level: K5 – Evaluate)

Problem Context – Modernising a Smart-City Network: A smart city will connect 2 million sensors, send each traffic-camera stream to several control rooms, and wants to change routing centrally.

Your Challenge: Write `2001:0db8:0000:0000:0000:ff00:0042:8329` in its shortest form, and build the EUI-64 interface identifier for a sensor with MAC address `00:1A:2B:3C:4D:5E` . Explain, using the TTL field, how `traceroute` uses ICMP Time Exceeded messages to discover each router. Find the multicast MAC address for the group `239.1.2.3` , and explain why one camera stream reaches several control rooms without being copied by the camera. An OpenFlow switch has the rules `(priority 200: dst 10.5.0.0/16 → port 3)` , `(priority 100: TCP port 80 → controller)` , `(priority 0: drop)` . Decide what happens to an HTTP packet to `10.5.1.1` and to an SSH packet to `10.6.1.1` , and judge whether SDN is worth adopting for the city.
Alternative Concept Analysis: Compare moving straight to IPv6 with running dual-stack or tunnelling IPv6 over IPv4 for the city. Calculate the extra header bytes of a 6in4 tunnel on a 100- byte sensor packet, and decide which approach the city should start with.

Evidence Expected: Correct IPv6, EUI-64 and multicast calculations, a correct traceroute explanation and a reasoned SDN decision.

DAY-4 DESIGN CHALLENGE: Q7 — Find the Weak Point, Redesign and Validate (13 Marks • Bloom's Level: K5 – Evaluate)

Review the evidence you generated in Q1–Q6. Select ONE network that shows a significant limitation as it grows or as the requirement changes. You are not permitted simply to add more equipment and claim the network is better. You must understand the limitation first and then develop an appropriate redesign.

A. Identify the Limitation (2 Marks): Using your own evidence, show where the design fails or slows down and which part of the addressing or routing design causes it.

B. Define the Design Objective (2 Marks): State a measurable objective (for example: convergence time, addresses wasted, routing-table size).

C. Design the Revised Network (4 Marks): Explain what you have changed and any new trade-off. Provide the topology diagram and address table.

D. Predict Before Implementing (1 Mark): Predict the improvement and the measure in which you expect it.

E. Implement and Validate (2 Marks): Compare Original → Redesigned in Packet Tracer using the same tests.

F. Defend the Final Design (2 Marks): Judge whether the objective was achieved, the new costs, and where the original may still be preferable.

Mark Distribution

Q1	IPv4 addressing, NAT and packet switching	K5 – Evaluate	6
Q2	Subnetting, VLSM and summarisation	K5 – Evaluate	8
Q3	Longest-prefix match and static routing	K5 – Evaluate	6
Q4	Distance-vector routing	K5 – Evaluate	6
Q5	Link-state routing (OSPF)	K5 – Evaluate	6
Q6	IPv6, ICMP, multicast and SDN	K5 – Evaluate	5
Q7	Day-4 Design Challenge – Redesign & Validate	K5 – Evaluate	13
TOTAL MARKS			50

Q1 — Spot the Invalid Addresses (Solution)

1.1 Subnet Analysis & Identification of the Two Traps

Device & Configured IP	Network Address	Broadcast Address	Usable Host Range	Valid Host? & Trap Diagnosis
Admission PC `172.16.45.200/20` GW: `172.16.32.1`	`172.16.32.0` (Block size 16 in 3rd octet: 32–47)	`172.16.47.255`	`172.16.32.1` – `172.16.47.254`	VALID HOST. Lies strictly inside `172.16.32.0/20` and shares the same subnet as gateway `172.16.32.1`.

Device & Configured IP	Network Address	Broadcast Address	Usable Host Range	Valid Host? & Trap Diagnosis
Lab Printer 10.0.15.255/21 GW: 10.0.8.1	10.0.8.0 (Block size 8 in 3rd octet: 8–15)	10.0.15.255	10.0.8.1 – 10.0.15.254	INVALID (TRAP 1 — Broadcast Address). Although .255 looks valid in a /21 , 10.0.15.255 is the directed broadcast address of 10.0.8.0/21 and cannot be assigned to a host interface.
X-ray Server 192.168.4.0/23 GW: 192.168.4.1	192.168.4.0 (Block size 2 in 3rd octet: 4–5)	192.168.5.255	192.168.4.1 – 192.168.5.254	INVALID (TRAP 2 — Network Address). While 192.168.5.0 is a valid host in /23 , 192.168.4.0 has all host bits equal to zero (the subnet ID itself) and is rejected by host IP stacks.
Clinic Router 203.0.113.10/29	203.0.113.8 (Block size 8 in 4th octet: 8–15)	203.0.113.15	203.0.113.9 – 203.0.113.14	VALID HOST. Second usable IP in 203.0.113.8/29 (6 usable public IPs total; hence 18 devices "ran out" without PAT/NAT).

1.2 PAT/NAT Port Disambiguation & Datagram vs. Virtual-Circuit Switching

How PAT Resolves Identical Source Ports (50000): Using Port Address Translation (PAT / NAPT), the clinic router translates both the private source IP and the TCP/UDP source port into unique external port numbers on its public IP 203.0.113.10 :

Protocol	Inside Local (Private IP:Port)	Inside Global (Public NAT IP:Port)	Outside Global (Server IP:Port)
TCP	192.168.1.10:50000 (PC 1)	203.0.113.10:40001	198.51.100.50:443
TCP	192.168.1.11:50000 (PC 2)	203.0.113.10:40002	198.51.100.50:443

When reply packets arrive at 203.0.113.10 , destination port 40001 maps back to 192.168.1.10:50000 while 40002 maps to 192.168.1.11:50000 , eliminating any collision.

Datagram vs. Virtual-Circuit Switching: In IP datagram switching, each packet is routed independently based on dynamic forwarding tables; if load balancing (ECMP) or a mid-transfer route change sends Packet 2 along a lower-latency path than Packet 1, Packet 2 arrives first (out-of-order delivery, reassembled by TCP). In virtual-circuit switching (e.g., MPLS-TE/ATM), a fixed path (VC identifier) is established during connection setup and every packet follows the exact same sequence of switches and links, guaranteeing in-order delivery at the cost of per-flow state and setup latency.

1.3 Alternative Concept Analysis: Public /27 vs. NAT behind /29

Address Count: 18 devices + 1 router gateway = 19 host addresses. A /27 provides $2^5 - 2 = 30$ usable public IPs (leaving 11 spare IPs; a /28 only gives 14 usable IPs, which is insufficient). Evaluation: Cost & Overhead: Leasing 32 public IPv4 addresses (/27) costs roughly 4× more monthly than a /29 (8 IPs), though NAT CPU/state overhead and ~50–150 µs translation delay are eliminated. Security & Exposure: Without NAT's implicit stateful filtering, all 18 clinic PCs become directly reachable from the public internet, requiring strict host/perimeter firewalling. Application Problems Eliminated: Protocols embedding IP addresses in payloads (SIP VoIP, active FTP, IPSec ESP without NAT-T) and peer-to-peer medical imaging connections work natively without Application Layer Gateways (ALGs) or port-forwarding rules.

Q2 — Subnet a Campus That Keeps Growing (Solution)

2.1 Equal-Size Subnetting vs. VLSM on 192.168.10.0/24

Method 1 — Equal-Size Subnetting: Four subnets are required, and the largest subnet (Labs) needs 100 + 2 = 102 addresses, which requires a /25 block (126 usable hosts). However, 192.168.10.0/24 only contains two /25 subnets (.0/25 and .128/25). If we split into four equal subnets (/26 , 62 usable hosts each), Staff (50), Library (25), and WAN (2) fit, but Labs (100 hosts) cannot fit without splitting Labs across two VLANs. Thus, equal-size subnetting fails to simultaneously provide 4 subnets and 100 contiguous hosts in a single /24 .

Method 2 — Variable-Length Subnet Masking (VLSM, Sorted Largest-First):

Department (Need)	Network / Prefix	Subnet Mask	First Usable Host	Last Usable Host	Broadcast
Labs (100 hosts)	192.168.10.0/25	255.255.255.128	192.168.10.1	192.168.10.126	192.168.10.127
Staff (50 hosts)	192.168.10.128/26	255.255.255.192	192.168.10.129	192.168.10.190	192.168.10.191
Library (25 hosts)	192.168.10.192/27	255.255.255.224	192.168.10.193	192.168.10.222	192.168.10.223
WAN Link (2 hosts)	192.168.10.224/30	255.255.255.252	192.168.10.225	192.168.10.226	192.168.10.227

The 60-Host Hostel Trap: Remaining free space is 192.168.10.228 – 192.168.10.255 (only 28 addresses, divisible into .228/30 , .232/29 , .240/28). A 60-host hostel requires a /26 block (64 addresses), so it cannot fit

without renumbering. Smallest change to make it fit: Total addresses needed = 128 + 64 + 64 + 32 + 4 = 292 > 256, which mathematically exceeds a single /24 ! Therefore, the smallest change is either (a) adding a second private block (e.g., 192.168.11.0/26) for the Hostel, or (b) expanding the campus supernet from 192.168.10.0/24 to 192.168.10.0/23 (512 addresses), placing Hostel at 192.168.11.0/26 with zero renumbering of existing VLANs.

2.2 Route Summarisation & Over-Summarisation Error

Converting the third octets of 172.16.8.0 – 172.16.11.0 to binary: 8 = 00001000 , 9 = 00001001 , 10 = 00001010 , 11 = 00001011 . The first 6 bits of the third octet (000010) are identical across all four subnets, giving prefix length 16 + 6 = 22. Thus, the exact summary route is 172.16.8.0/22 . If 172.16.12.0/24 (00001100) is added and summarised as 172.16.8.0/21 , that route covers 172.16.8.0 through 172.16.15.255 —causing a blackhole / route hijack for 172.16.13.0/24 , 172.16.14.0/24 , and 172.16.15.0/24 if those subnets reside elsewhere.

2.3 Alternative Concept Analysis: Smallest-First VLSM Allocation

Assigning smallest-first allocates WAN at .0/30 (0–3), then Library (32 block) must align on a multiple of 32 (so .4 – .31 is wasted or overlaps if forced at .4/27 , which actually resolves to .0/27 and collides with WAN!). Similarly, Staff (64 block) at .64/26 and Labs (128 block) at .128/25 leave .4 – .63 fragmented into unusable sub-blocks. Assigning largest blocks first guarantees natural binary boundary alignment with zero fragmentation.

Q3 — Which Way Will the Packet Go? (Solution)

3.1 Longest-Prefix Match Evaluation

Routers forward packets using the Longest-Prefix Match (LPM) rule: among all routing-table entries whose network prefix matches the destination IP, select the entry with the largest subnet mask length (/n):

Destination IP	All Matching Routes in Table	Winning Route (Longest Prefix)	Next Hop
10.1.2.200	/8 , /16 , /24 , /25 (.128 – .255), /0	10.1.2.128/25 (25 bits)	192.0.2.13
10.1.2.100	/8 , /16 , /24 (.0 – .255), /0 (not /25)	10.1.2.0/24 (24 bits)	192.0.2.9
10.1.9.9	10.0.0.0/8 , 10.1.0.0/16 , 0.0.0.0/0	10.1.0.0/16 (16 bits)	192.0.2.5
10.7.7.7	10.0.0.0/8 , 0.0.0.0/0	10.0.0.0/8 (8 bits)	192.0.2.1
8.8.8.8	0.0.0.0/0 (Default route only)	0.0.0.0/0 (0 bits)	198.51.100.1

3.2 Floating Static Route & DCE Clock Rate Behaviour

Adding a backup link (e.g., 10.0.0.4/30 via 10.0.0.6) with ip route 192.168.20.0 255.255.255.0 10.0.0.6 200 keeps the backup route out of the RIB while the primary static route (AD = 1) is active. When Serial0/0/0 is shut down, R1 withdraws the AD 1 route and immediately installs the AD 200 floating static route. Removing

DCE Clock Rate: Issuing `no clock rate 64000` on the DCE interface removes the physical-layer timing signal, causing the serial line protocol to transition from `up/up` to `up/down` and dropping the connected `10.0.0.0/30` route.

3.3 Alternative Concept Analysis: Static Routing vs. OSPF

For 2 campuses, static routing requires 1 route line per router (2 total) vs. ~4 OSPF configuration lines per router (8 total). However, for 5 campuses in a mesh, static routing requires 5 × 4 = 20 manual route statements (plus 20 backup statements = 40 lines), whereas OSPF still requires only ~4 lines per router (20 total) and automatically discovers all topologies. On link failure, a direct interface-down triggers floating static failover in <50 ms, whereas default OSPF (Hello 10s / Dead 40s) takes 40–45 seconds unless tuned with BFD or fast timers, though OSPF handles indirect upstream failures that static routes cannot detect without IP SLA.

Q4 — Watch Distance-Vector Routing Count to Infinity (Solution)

4.1 Round-by-Round Bellman-Ford Convergence (5 Routers)

Link costs: A–B=2, A–C=7, B–C=3, B–D=8, C–D=1, D–E=4. Initially (Round 0), routers know only direct neighbours:

Round	Router A (A,B,C,D,E)	Router B (A,B,C,D,E)	Router C (A,B,C,D,E)	Router D (A,B,C,D,E)	Router E (A,B,C,D,E)
Init (0)	0, 2, 7, 99, 99	2, 0, 3, 8, 99	7, 3, 0, 1, 99	99, 8, 1, 0, 4	99, 99, 99, 4, 0
Round 1	0, 2, 5, 10, 99	2, 0, 3, 4, 12	5, 3, 0, 1, 5	8, 4, 1, 0, 4	99, 12, 5, 4, 0
Round 2	0, 2, 5, 6, 14	2, 0, 3, 4, 8	5, 3, 0, 1, 5	6, 4, 1, 0, 4	12, 8, 5, 4, 0
Round 3	0, 2, 5, 6, 10	2, 0, 3, 4, 8	5, 3, 0, 1, 5	6, 4, 1, 0, 4	10, 8, 5, 4, 0
Round 4	0, 2, 5, 6, 10	2, 0, 3, 4, 8	5, 3, 0, 1, 5	6, 4, 1, 0, 4	10, 8, 5, 4, 0 (No change → Stop)

Convergence: Required 4 synchronous rounds (3 update rounds + 1 confirmation round). Why A→C is not direct: The direct A–C link costs 7, whereas the two-hop path `A → B → C` costs 2 + 3 = 5, which Bellman-Ford strictly prefers. After Link C–D Fails (cost = 99): Re-running from scratch converges in 4 rounds to: A: `0, 2, 5, 10, 14` ; B: `2, 0, 3, 8, 12` ; C: `5, 3, 0, 11, 15` ; D: `10, 8, 11, 0, 4` ; E: `14, 12, 15, 4, 0` (traffic reroutes via B–D cost 8).

4.2 Count-to-Infinity on `X — Y — Z` & Split Horizon with Poison Reverse

Before failure, $D_Y(Z) = 1$ (direct) and $D_X(Z) = 2$ (via Y). When Y–Z fails ($c(Y,Z) = \infty$): • Without Split Horizon: Round 1: Y hears $D_X(Z)=2$ from X and sets $D_Y(Z) = 1 + 2 = 3$. Round 2: X hears $D_Y(Z)=3$ and updates $D_X(Z) = 1 + 3 = 4$. Round 3: Y updates to 5, continuing until ∞ (16 in RIP).

• With Split Horizon + Poison Reverse: Because X routes to Z through Y, X advertises $D_X(Z) = \infty$

(poisoned metric 16) back to Y. When Y–Z fails in Round 1, Y sees c(Y,Z)=∞ and X's poisoned $D_X(Z)=\infty$, immediately setting $D_Y(Z) = \infty$ in 1 round and notifying X in Round 2 (loop-free convergence in 2 rounds).

4.3 Alternative Concept Analysis: Link-State vs. Distance-Vector

When C–D fails in Link-State (OSPF), C and D flood a single updated Link-State Advertisement (LSA) across all 6 links (~12 LSA packets total), and every router independently runs Dijkstra locally to converge in 1 flooding diameter (2 hops / <1 second). Distance-vector exchanges full vectors iteratively over multiple 30-second periodic intervals (RIP) and risks transient loops during multi hop failures.

Q5 — Why Does OSPF Treat 1 Gbps and 10 Gbps the Same? (Solution)

5.1 Default Reference Bandwidth Problem & Recalculation

OSPF calculates interface cost as $\text{Cost} = \lfloor \text{Reference Bandwidth} / \text{Interface Bandwidth} \rfloor$, floored at a minimum integer value of 1. With Cisco's legacy default reference bandwidth of 100 Mbps (10^8 bps):

Link Speed	Default Cost (Ref = 100 Mbps)	Revised Cost (Ref = 100 Gbps = 100,000 Mbps)
10 Mbps	100 / 10 = 10	100,000 / 10 = 10,000
100 Mbps	100 / 100 = 1	100,000 / 100 = 1,000
1 Gbps (1,000 Mbps)	$\lfloor 100 / 1000 \rfloor \to \mathbf{1}$	100,000 / 1,000 = $\mathbf{100}$
10 Gbps (10,000 Mbps)	$\lfloor 100 / 10000 \rfloor \to \mathbf{1}$	100,000 / 10,000 = $\mathbf{10}$

Problem: Both 1 Gbps and 10 Gbps truncate to cost 1, making OSPF treat them as equal-cost paths (ECMP) or prefer a 1-hop 1 Gbps link over a 2-hop 10 Gbps backbone. Setting `auto-cost reference bandwidth 100000` (100 Gbps) cleanly separates all four speeds (10,000 / 1,000 / 100 / 10) and leaves headroom for 40G/100G upgrades.

5.2 Hand-Worked Dijkstra Shortest-Path Tree (6-Router ISP Topology)

Topology links (with Ref = 100 Gbps): R1–R2 (1 Gbps = 100), R1–R3 (10 Gbps = 10), R3–R4 (10 Gbps = 10), R4–R2 (10 Gbps = 10), R2–R5 (10 Gbps = 10), R4–R6 (1 Gbps = 100), R5–R6 (10 Gbps = 10). Running Dijkstra from source R1:

Step 1 (Visit R1, dist=0): Relax neighbours → $d(\text{R2})=100$ (via R1), $d(\text{R3})=10$ (via R1).

Step 2 (Visit R3, dist=10): Relax R4 → $d(\text{R4}) = 10 + 10 = 20$ (via R3). Step 3 (Visit R4, dist=20): Relax R2 → $20 + 10 = 30 < 100$, update $d(\text{R2})=30$ (via R4); relax R6 → $d(\text{R6}) = 20 + 100 = 120$ (via R4).

Step 4 (Visit R2, dist=30): Relax R5 → $d(\text{R5}) = 30 + 10 = 40$ (via R2). Step 5 (Visit R5, dist=40): Relax R6 → $40 + 10 = 50 < 120$, update $d(\text{R6})=50$ (via R5). Step 6 (Visit R6, dist=50): Complete.

Path R1→R2 uses the 3-hop 10 Gbps ring (`R1→R3→R4→R2` , cost 30) instead of the direct 1 Gbps link (cost 100).

Single vs. Multi-Area Justification: For a 6-router ISP core, a single Area 0 (Backbone) is optimal: LSDB size is tiny (<20 LSAs), SPF runs in <1 ms, and multi-area ABRs would block end-to end inter-area shortest-path visibility and add unnecessary configuration complexity. Multi-area OSPF is only warranted above ~50–80 routers.

5.3 Alternative Concept Analysis: EIGRP & IS-IS

IS-IS defaults to a narrow metric of 10 on every interface regardless of speed, suffering the exact same problem unless wide metrics are manually configured. EIGRP uses \text{Metric} = 256 \times (10^7 / \text{BW}_{\text{kbps}} + \text{Delay}/10): for 1 Gbps (10^6 kbps, delay 10 µs), \text{Metric} = 256 \times (10 + 1) = \mathbf{2,816}; for 10 Gbps (10^7 kbps, delay 10 µs), \text{Metric} = 256 \times (1 + 1) = \mathbf{512}. Thus, EIGRP automatically distinguishes 1 Gbps from 10 Gbps out of the box without changing any reference bandwidth.

Q6 — IPv6, ICMP, Multicast and SDN in a Smart City (Solution)

6.1 IPv6 Compression, EUI-64, Traceroute & Multicast MAC

Shortest IPv6 Form: `2001:0db8:0000:0000:0000:ff00:0042:8329` → strip leading zeros per hextet and compress the longest run of all-zero hextets with `::` → `2001:db8::ff00:42:8329` . EUI-64 Interface ID from MAC `00:1A:2B:3C:4D:5E` : Split MAC in half and insert `FF:FE` → `00:1A:2B:FF:FE:3C:4D:5E` . Flip the 7th (Universal/Local) bit of the first byte: `0x00` (`00000000`) → `0x02` (`00000010`). Result: `021a:2bff:fe3c:4d5e` .

How `traceroute` Uses TTL & ICMP: Traceroute sends probe packets starting with IP `TTL = 1` . Router 1 decrements TTL to 0, drops the packet, and returns an ICMP Time Exceeded (Type 11, Code 0) message revealing Router 1's IP. Traceroute increments TTL to 2, 3, … discovering each hop in sequence until reaching the destination (which replies with ICMP Echo Reply or Port Unreachable).

Multicast MAC for `239.1.2.3` : Map the low-order 23 bits of the IPv4 multicast address onto the IEEE OUI prefix `01:00:5E` . `239.1.2.3` in hex is `EF.01.02.03` ; masking the second octet with `0x7F` keeps `01` , yielding `01:00:5E:01:02:03` . One camera stream reaches multiple control rooms without camera duplication because multicast-enabled routers (PIM/IGMP) replicate frames at branching downstream switches only along branches with active IGMP group subscribers.

6.2 OpenFlow Rule Evaluation & SDN Judgement

HTTP packet to `10.5.1.1` (TCP dst port 80): Matches both Priority 200 (`dst 10.5.0.0/16`) and Priority 100 (`TCP port 80`). OpenFlow always executes the highest-priority matching rule → Forwarded out Port 3 (never sent to controller).

SSH packet to `10.6.1.1` (TCP dst port 22): Does not match `10.5.0.0/16` or TCP port 80; matches Priority 0 catch-all → Dropped.

SDN Judgement: Worth adopting for the smart-city core/aggregation network because centralized OpenFlow controllers allow instant dynamic rerouting of emergency video feeds, per service QoS slicing, and automated ACL updates across thousands of switches, provided redundant controllers (cluster of 3) are deployed to avoid a single point of failure.

6.3 Alternative Concept Analysis: Native IPv6 vs. Dual-Stack vs. 6in4 Tunnelling

A 6in4 tunnel encapsulates an IPv6 packet inside a 20-byte IPv4 header (adding 20 bytes overhead, a 20% payload overhead on a 100-byte sensor packet, reducing effective throughput and risking MTU

fragmentation). Moving straight to native IPv6-only on the new 2-million-sensor IoT network (with NAT64/DNS64 at the edge for legacy IPv4 control servers) avoids 6in4 header overhead and eliminates maintaining two routing tables (dual-stack) across 2 million nodes.

DAY-4 DESIGN CHALLENGE: Q7 — Find the Weak Point, Redesign and Validate (Solution)

A. Identify the Limitation (2 Marks)

Selecting the Q2 Engineering College Campus Network (`192.168.10.0/24`): As proved in Q2, our initial VLSM design exhausted 228 of 256 addresses across Labs (`/25`), Staff (`/26`), Library (`/27`), and WAN (`/30`), leaving only 28 fragmented addresses (`.228` – `.255`). When the college expands by adding a 60-host Hostel (and future student labs), the single `/24` address space suffers hard exhaustion, and the flat Layer-2/single-router topology lacks inter-VLAN isolation and summarisable growth headroom.

B. Define the Design Objective (2 Marks)

1. Address Capacity & Efficiency: Expand campus address space to accommodate the 60-host Hostel plus at least 100% future growth headroom (>200 spare host IPs) while keeping internal address utilisation above 55% and zero renumbering of existing subnets.
2. Routing Table Compactness: Aggregate all internal campus subnets into 1 single summary route (`/23`) advertised to the ISP/WAN router (reducing external routing table entries by 80%, from 5 prefixes to 1).

C. Design the Revised Network (4 Marks)

We re-architect the campus around a hierarchical `192.168.10.0/23` supernet (512 addresses: `192.168.10.0` – `192.168.11.255`) with a Layer-3 distribution switch (Router-on-a-Stick / SVI core) and a single summary route `192.168.10.0/23` toward the WAN border router. Trade-off: adds VLAN trunking/ACL configuration complexity on the core switch in exchange for clean growth and route aggregation.

VLAN / Segment	Hosts Needed	Redesigned Subnet	Usable Host Range	Broadcast
VLAN 10 — Labs (Existing)	100	192.168.10.0/25	.10.1 – .10.126	192.168.10.127
VLAN 20 — Staff (Existing)	50	192.168.10.128/26	.10.129 – .10.190	192.168.10.191
VLAN 30 — Library (Existing)	25	192.168.10.192/27	.10.193 – .10.222	192.168.10.223
WAN Point-to-Point (Existing)	2	192.168.10.224/30	.10.225 – .10.226	192.168.10.227
VLAN 40 — Hostel (New)	60	192.168.11.0/26	.11.1 – .11.62	192.168.11.63

Reserved Expansion Pool	190 spare	192.168.11.64/26 + .128/25	.11.65 – .11.254	—

D. Predict Before Implementing (1 Mark)

We predict that: (1) all 60 Hostel hosts will obtain valid IPs and achieve 100% ping reachability to Labs, Staff, Library, and WAN without touching any existing device IP; (2) free contiguous host capacity will rise from 24 usable IPs to 214 usable IPs; and (3) the WAN edge router's routing table will shrink from 5 specific campus routes to 1 summary route (192.168.10.0/23).

E. Implement and Validate (2 Marks)

Validation Metric (Packet Tracer Test Suite)	Original Design (/24)	Redesigned Network (/23)
Hostel (60-host) Allocation & Ping Test	FAIL (Insufficient space; 0% reachable)	PASS (100% ICMP echo success, avg 2 ms)
Existing Subnets Renumbered	N/A (Blocked)	0 devices renumbered
Remaining Usable Host Capacity	24 IPs (fragmented)	214 IPs (contiguous /26 + /25)
WAN Router Campus Prefix Count	4 (5 with Hostel)	1 summary route (192.168.10.0/23)

F. Defend the Final Design (2 Marks)

Conclusion: Both measurable objectives were fully achieved: the 60-host Hostel integrated seamlessly with zero renumbering, spare capacity grew nearly 9×, and WAN routing state dropped by 80%. New Costs & Trade-Offs: Requires coordinating a /23 allocation from the upstream enterprise/ISP block and enforcing inter-VLAN ACLs so Hostel traffic cannot access Staff/Admin servers. When Original Is Preferable: For a static branch office with fewer than 180 fixed hosts and zero projected expansion, the original single /24 VLSM plan remains simpler and conserves upstream IPv4 space.